D N-shot Prompt (examples selected randomly from GSM8k train)

Below is an example prompt. For each question, we select five random examples from GSM8k to use as n-shot examples, which vary for each new question from the GSM1k/GSM8k test set. While evaluation methods vary between models, this is the most common approach to evaluating GSM8k.

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Question: Jen and Tyler are gymnasts practicing flips. Jen is practicing the triple-flip
while Tyler is practicing the double-flip. Jen did sixteen triple-flips during practice.
Tyler flipped in the air half the number of times Jen did. How many double-flips did Tyler do?
Answer: Jen did 16 triple-flips, so she did 16 * 3 = <<16*3=48>>48 flips.
Tyler did half the number of flips, so he did 48 / 2 = <<48/2=24>>24 flips.
A double flip has two flips, so Tyler did 24 / 2 = <<24/2=12>>12 double-flips.
#### 12
Question: Four people in a law firm are planning a party. Mary will buy a platter of pasta
for $20 and a loaf of bread for $2. Elle and Andrea will split the cost for buying 4 cans
of soda which cost $1.50 each, and chicken wings for $10. Joe will buy a cake that costs
$5. How much more will Mary spend than the rest of the firm put together?
Answer: Mary will spend $20 + $2 = $<<20+2=22>>22.
Elle and Andrea will spend $1.5 x 4 = $<<1.5*4=6>>6 for the soda.
Elle and Andrea will spend $6 + $10 = $<<6+10=16>>16 for the soda and chicken wings.
Elle, Andrea, and Joe together will spend $16 + $5 = $<<16+5=21>>21.
So, Mary will spend $22 - $21 = $<<22-21=1>>1 more than all of them combined.
#### 1
Question: A charcoal grill burns fifteen coals to ash every twenty minutes of grilling.
The grill ran for long enough to burn three bags of coals. Each bag of coal contains 60
coals. How long did the grill run?
Answer: The grill burned 3 * 60 = <<3*60=180>>180 coals.
It takes 20 minutes to burn 15 coals, so the grill ran for 180 / 15 * 20 =
<<180/15*20=240>>240 minutes.
#### 240
Question: A bear is preparing to hibernate for the winter and needs to gain 1000 pounds.
At the end of summer, the bear feasts on berries and small woodland animals. During autumn,
it devours acorns and salmon. It gained a fifth of the weight it needed from berries during
summer, and during autumn, it gained twice that amount from acorns. Salmon made up half of
the remaining weight it had needed to gain. How many pounds did it gain eating small animals?
Answer: The bear gained 1 / 5 * 1000 = <<1/5*1000=200>>200 pounds from berries.
It gained 2 * 200 = <<2*200=400>>400 pounds from acorns.
It still needed 1000 - 200 - 400 = <<1000-200-400=400>>400 pounds.
Thus, it gained 400 / 2 = <<400/2=200>>200 pounds from salmon.
Therefore, the bear gained 400 - 200 = <<400-200=200>>200 pounds from small animals.
#### 200
Question: Brendan can cut 8 yards of grass per day, he bought a lawnmower and it helped
him to cut more yards by Fifty percent per day. How many yards will Brendan be able to cut
after a week?
Answer: The additional yard Brendan can cut after buying the lawnmower is 8 x 0.50 =
<<8*0.50=4>>4 yards.
So, the total yards he can cut with the lawnmower is 8 + 4 = <<8+4=12>>12.
Therefore, the total number of yards he can cut in a week is 12 x 7 = <<12*7=84>>84 yards.
#### 84
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E Results with an Alternative Prompt

As an ablation, we evaluate all models with an alternative prompt scheme and compare results with our primary findings. This prompt is available under the LM Evaluation Harness as a "chain-of-thought" prompt. However, manually examining the prompt (provided in full below) reveals that the primary difference with the standard n-shot prompt lies not in chain-of-thought reasoning but rather using a set of non-GSM8k problems as guiding examples as well as providing an alternative answer format. We choose to use the standard prompt to match typical evaluation methods widespread in the field but also report these results for completeness.

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Q: There are 15 trees in the grove. Grove workers will plant trees in the
     grove today. After they are done, there will be 21 trees. How many
    trees did the grove workers plant today?
A: There are 15 trees originally. Then there were 21 trees after some
    more were planted. So there must have been 21 − 15 = 6. The answer is
     6.

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how
    many cars are in the parking lot?
A: There are originally 3 cars. 2 more cars arrive. 3 + 2 = 5. The answer
     is 5.

Q: Leah had 32 chocolates and her sister had 42. If they ate 35, how many
     pieces do they have left in total?
A: Originally, Leah had 32 chocolates. Her sister had 42. So in total
    they had 32 + 42 = 74. After eating 35, they had 74 − 35 = 39. The
    answer is 39.

Q: Jason had 20 lollipops. He gave Denny some lollipops. Now Jason has 12
     lollipops. How many lollipops did Jason give to Denny?
A: Jason started with 20 lollipops. Then he had 12 after giving some to
    Denny. So he gave Denny 20 − 12 = 8. The answer is 8.

Q: Shawn has five toys. For Christmas, he got two toys each from his mom
    and dad. How many toys does he have now?
A: Shawn started with 5 toys. If he got 2 toys each from his mom and dad,
     then that is 4 more toys. 5 + 4 = 9. The answer is 9.

Q: There were nine computers in the server room. Five more computers were
     installed each day, from monday to thursday. How many computers are
    now in the server room?
A: There were originally 9 computers. For each of 4 days, 5 more
    computers were added. So 5 ∗ 4 = 20 computers were added. 9 + 20 is
    29. The answer is 29.

Q: Michael had 58 golf balls. On tuesday, he lost 23 golf balls. On
    wednesday, he lost 2 more. How many golf balls did he have at the end
     of wednesday?
A: Michael started with 58 golf balls. After losing 23 on tuesday, he had
     58 − 23 = 35. After losing 2 more, he had 35 − 2 = 33 golf balls.
    The answer is 33.

Q: Olivia has $23. She bought five bagels for $3 each. How much money
    does she have left?
A: Olivia had 23 dollars. 5 bagels for 3 dollars each will be 5 x 3 = 15
    dollars. So she has 23 − 15 dollars left. 23 − 15 is 8. The answer is
     8.
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We report our results in Table F. While there is significant variance based on prompt, the general trend of which model families are overfit is similar.

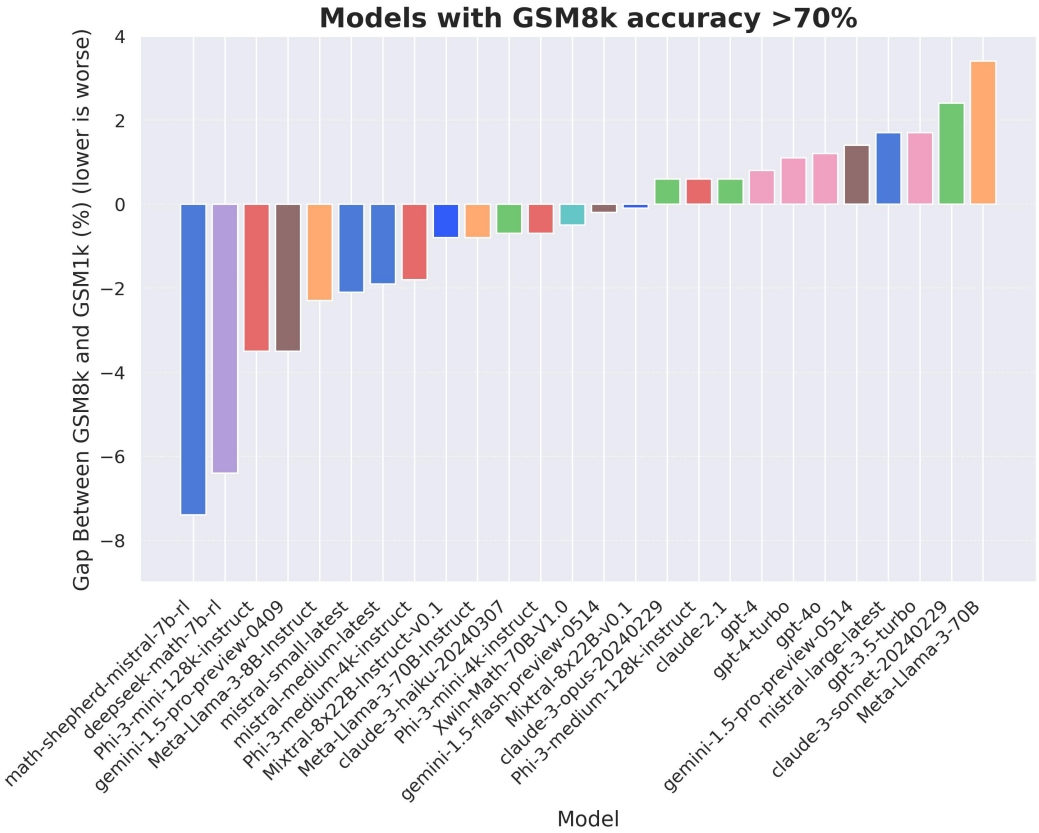

Figure 7: Gap in accuracy between GSM8k and GSM1k for models that score above 70% on GSM8k.